

Introduction to Thesis - Senior Project

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Abstract

The increasing demand for efficient and flexible computation in digital signal processing (DSP) applications has driven the exploration of custom processor architectures. This project focuses on the design and implementation of a 32-bit RISC-V softcore central processing unit (CPU) optimized for DSP operations. The objective is to develop a processor that maintains the simplicity and modularity of the RISC-V instruction set architecture (ISA) while introducing hardware-level enhancements to accelerate DSP workloads such as filtering, convolution, and fast Fourier transforms (FFT). The softcore is implemented on a field-programmable gate array (FPGA) platform, allowing for rapid prototyping and performance evaluation. Custom instructions and pipeline optimizations are integrated to improve throughput and reduce latency for fixed-point arithmetic operations common in DSP tasks. The resulting architecture demonstrates how open-source RISC-V designs can be adapted to meet domain-specific performance requirements, bridging the gap between general-purpose computation and dedicated signal processing hardware.

1 Introduction

For this project, we were set to find a problem and solve it. We wanted to focus on designing a CPU that is specifically suited to solving digital signal processing (DSP) calculations. That is when we came up with the idea of creating and designing a 32-Bit softcore RISC-V CPU that specializes in DSP calculations. To be able to do this, we first needed to come

up with a plan on how we are actually going to proceed to accomplish this.

1.1 Research

We first needed to begin doing research to come up on how we could accomplish this or if it was even possible. As of this moment we have read 8+ main articles relating to our project. Many composing of dot product convolution, Multiply-Accumulate unit, and block diagram for DSP CPU architecture. Many of these articles are necessary for us to understand and know what is needed to accomplish this task of ours. We believe that we have a pretty good grasp of what to expect and is needed and will begin laying down the foundation of actual implementation and design of our project.

1.2 Roadmap

The first step to our road map of researching the components of our project we believe now is complete. We have also researched and laid the preparation work of FPGA board that we will working on to implement the soft core processor on. The next step is to design a block diagram to get a sense of how our CPU architecture will look like. The main components would be the Arithmetic Logic Unit (ALU), program counter, registers, etc. Once this is complete we will begin to code our soft core processor using verily as our main language to be able to do this. Software that we will be using to be able to utilized our soft core is Xilinx Vivado to be able to communicate with our FPGA board. More research will be needed on how to program and communicate using Vivado with our FPGA board. For example assign-

ing pins onto the board, running code, and connecting all our main components to replicate our block diagram. Once this stage is complete, we will then move onto our testing phase where we will be feeding our soft core calculations and will be comparing the performance metrics with a different processor. Further research will be needed as we are unsure of how to test bench our processor at the moment.

1.3 Why RISC-V? Why 32-Bit?

Why are we using RISC-V when we could instead use x86 or ARM? The reason is because with RISC-V it is free and open source while with x86 and ARM, you would need to pay a licensing fee to be able to use them. Another reason is that RISC-V is designed in a way for small groups to easily be able to use and design with. It is also very much used for academic use which is especially useful in our case for our project. So then why are we deciding to go with 32-bit and not 64-bit? The main reason is 32-bit is the most feasible one we believe that we can accomplish. 64-bit is just too ambitious and would not be possible with our circumstances. 16-bit on the other hand is the opposite. It is too simple and not ambitious enough with our project which makes 32-bit the perfect sweet spot.

1.4 Related Work

In our early research, there was a similar project done called *RISC-v BOOM* where it shows a walk through of how they were able to create a 32-bit RISC-V processor. BOOM stands for Berkeley out-of-order RISC-V Processor. With theirs they had a block diagram, pipeline, different type of stages, and much more. With this similar work, we are able to use this as a reference for our project and to know what to expect. Of course for our project, it will not exactly be like the BOOM processor. The key differences with our project and BOOM project is with ours we set to add a MAC unit as well be tailored for DSP which the BOOM does not.

1.5 Challenges

There have been a few challenges we have had to inevitably our project would have to face. One such from the very beginning was a budget of 200-300 dollars for our project. However, since our project does not require a whole set list of materials this was quickly overcome. The only problem we face is choosing what FPGA board we were going to need in order to design our soft core since it would not be possible to get a actual hard physical processor for our design. We opted to go for the *FPGA Board Pros and Cons Digilent Arty A7-100T*. We believe this board would best be suited to be used in our case for our project. It is an AMD board meaning we would have to use their software when implementing. Another challenge we have faced was deciding what exactly we wanted our processor to do. We knew that we wanted to design 32-Bit processor but we were not sure exactly for what purpose for. That is when we came up with the idea of creating a processor specialized for computing DSP calculations. We have faced other challenges as well however, there were minor compared to the ones discussed.

2 Conclusion

In conclusion, this project aims to design and implement a 32-bit RISC-V softcore CPU tailored for digital signal processing (DSP) applications. Through our research, we have gained a strong understanding of the core concepts required to achieve this, such as Multiply-Accumulate (MAC) units, convolution operations, and DSP-specific architectural optimizations. The RISC-V architecture was selected for its open-source nature, flexibility, and suitability for academic and experimental use, allowing us to focus on innovation rather than licensing or proprietary limitations.

We have established a clear roadmap that begins with architectural design through block diagrams and continues toward hardware description in Verilog, synthesis using Xilinx Vivado, and deployment on an FPGA platform—specifically the Digilent Arty A7-100T board. Although challenges such as hardware

selection, communication setup, and testing methodologies have arisen, each has provided valuable insight into practical CPU design and implementation.

As we proceed with development, our next steps involve finalizing the CPU architecture, implementing custom DSP-oriented instructions, and validating performance through benchmarking against existing processors. Ultimately, this project will demonstrate how a customizable RISC-V design can be effectively adapted to accelerate DSP workloads, contributing both to the academic understanding and practical advancement of domain-specific processor design.